

Tokenization Schemes for ODE Transformers: Learnability and Physical Fidelity on the Lorenz System

Sophia, Bob, Yuhan, Princeton
Math 270A

June 2026

Abstract

For scientific autoregressive modeling, tokenization is often treated as a standardized preprocessing step, yet for continuous dynamical systems it is fundamentally a modeling choice. Machine learning models operate on discrete token sequences, whereas dynamical systems evolve in a continuous state space. As a result, the method used to convert trajectories into tokens plays a central role in determining what information is preserved and how effectively the dynamics can be learned. In this work, we compare five tokenization schemes for the Lorenz system: uniform coordinate binning, quantile binning, K-means clustering, hierarchical quantization, and residual vector quantization (RVQ). We conduct an ablation study across these schemes while keeping the dataset, Transformer architecture, and training pipeline fixed. Each scheme is evaluated using reconstruction fidelity, next-token prediction accuracy, normalized learnability in bits per physical timestep, and free-running rollout quality measured by marginal Wasserstein distance. No single tokenizer dominates across all metrics: hierarchical quantization achieves the best reconstruction fidelity, whereas K-means produces the most learnable representation and the closest match to the true attractor distribution. These results suggest that, for chaotic systems, tokenization should be judged not only by predictive performance but also by how well it preserves the underlying geometry and long-term dynamics.

1 Introduction

The application of modern sequence models to scientific computing poses a fundamental representational challenge. Physical systems typically evolve in continuous state spaces governed by differential equations, while autoregressive architectures typically operate on sequences of discrete tokens. To bridge this gap, a discretization step, also known as tokenization, is introduced. Though tokenization is usually treated as incidental preprocessing, in practice, it is a consequential modeling decision. Mapping a continuous trajectory onto a token sequence uniquely determines which dynamical features are preserved, which are lost, and how naturally the resulting sequences can be learned by an autoregressive model. Our goal is to investigate the variations between such mappings, using Lorenz system trajectory prediction as our autoregressive task.

The Lorenz system provides an ideal setting for such an investigation [1]. Its strange attractor is geometrically rich yet low-dimensional: trajectories are bounded, never periodic, and exponentially sensitive to initial conditions. Because of this sensitivity, exact long-horizon state prediction is fundamentally unreliable, even for a model that has correctly internalized the qualitative dynamics. A more meaningful criterion for evaluating a learned model is therefore not pointwise trajectory accuracy but rather whether free-running rollouts remain on the attractor and reproduce its

distributional structure. This shift in evaluation philosophy is central to the conclusions of the present work.

The autoregressive Transformer [4] is a natural architecture for this setting: given a sequence of tokens encoding a trajectory prefix, the model is trained to predict the next token. Before such a model can be applied, however, each continuous state $\mathbf{x}_t = (x_t, y_t, z_t)$ must be mapped to one or more discrete tokens, and the choice of this mapping is precisely what the present ablation study examines. Five schemes are compared: uniform coordinate binning, quantile binning, K-means clustering, hierarchical quantization, and residual vector quantization, while the dataset, architecture, and training pipeline are held fixed, so that any differences in performance can be attributed to the tokenization alone.

The central finding is that the ranking across tokenizers depends on which scientific goal is prioritized. Hierarchical quantization best preserves the continuous state, achieving the lowest reconstruction error by representing each coordinate at two levels of resolution. K-means clustering, by contrast, operates directly on the joint state-space geometry, producing a compact single-token-per-timestep representation that is substantially easier for the Transformer to learn and that yields the closest distributional match to the true attractor in free-running rollouts. These outcomes are not in tension with each other; they reflect the fact that reconstruction fidelity and dynamical learnability are genuinely distinct objectives, and that the most appropriate tokenizer depends on what the trained model is ultimately intended to do.

2 Scientific Formulation

2.1 Lorenz dynamics

We define our Lorenz system as follows:

$$\dot{x} = \sigma(y - x), \tag{1}$$

$$\dot{y} = x(\rho - z) - y, \tag{2}$$

$$\dot{z} = xy - \beta z, \tag{3}$$

with the classical parameter values

$$\sigma = 10, \quad \rho = 28, \quad \beta = \frac{8}{3}.$$

For these parameters, trajectories approach a *strange* attractor, meaning they stay bounded, they never settle into a periodic orbit, and they are sensitive to initial conditions. Figure 1 shows examples of the resulting attractor geometry and coordinate time series.

The scientific structure of the problem affects evaluation. If a model is rolled forward autoregressively, small errors in state or token choice can move the predicted trajectory away from the true path. In a chaotic system, this is expected. The more important question is whether the generated path remains on or near the attractor and whether its distribution resembles the true test trajectories.

2.2 Data generation and standardization

Our dataset contains 68 total Lorenz trajectories, 60 of which are for training and 8 for testing. Each trajectory contains 1,500 post-transient timesteps. Data were generated by RK-45 integration

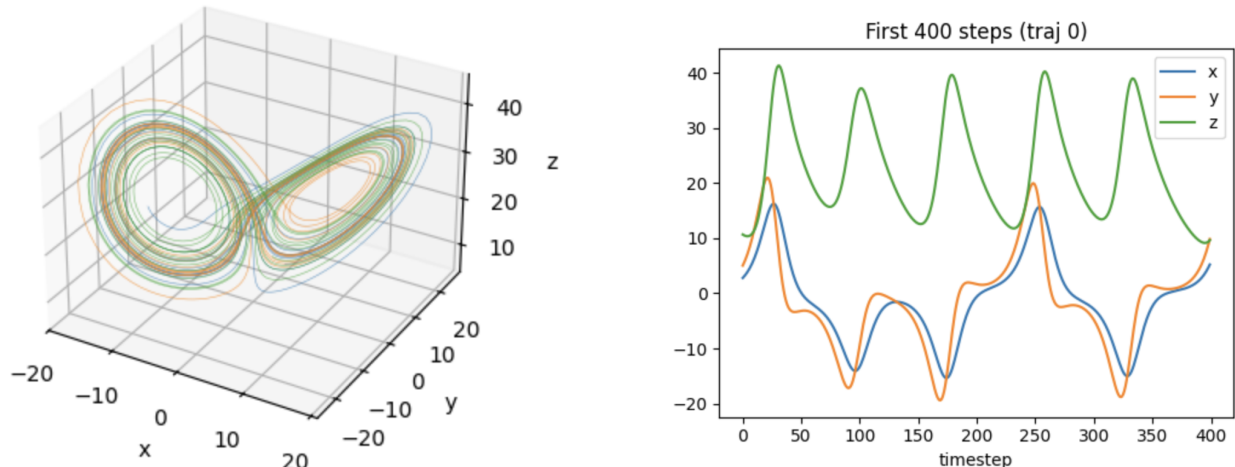


Figure 1: Example of a Lorenz Trajectory.

with timestep $\Delta t = 0.01$, discarding an initial 500-step transient so that the retained states lie on the attractor. The train and held-out trajectories use the same Lorenz parameters but different initial conditions.

Before tokenization, each coordinate is standardized using statistics fit on the training set only:

$$\tilde{v} = \frac{v - \mu_v}{\sigma_v}, \quad v \in \{x, y, z\}.$$

This places the three coordinates on a comparable scale and avoids leaking held-out trajectory information into the tokenizer fit.

3 Methodology

3.1 Controlled ablation design

The experiment is an ablation study over tokenizers. The Lorenz simulation, train/test split, standardization procedure, Transformer objective, and evaluation metrics are held fixed. Only the tokenization scheme changes.

Training windows are sampled within individual trajectories, not across concatenated trajectory boundaries. This preserves physical time. Crossing from the end of one trajectory into the beginning of another would create artificial transitions that do not correspond to the ODE flow.

3.2 Tokenization schemes

Table 1 summarizes the five tokenizers. For the coordinate-wise tokenizers, the implementation emits one coordinate token for each of x , y , and z , so the sequence contains three tokens per physical timestep. Equivalently, the triple (b_x, b_y, b_z) can be packed into a single composite state index for indexing purposes, but the reported vocabulary size is the per-coordinate token alphabet used in the token stream. All tokenizers are fit exclusively on training trajectories; the learned parameters

(bin edges, cluster centers, or codebook entries) are then applied to both train and test data without re-fitting.

The vocabulary sizes and tokens-per-step for each scheme were chosen to balance representational capacity against sequence complexity under a fixed Transformer probe. For coordinate-wise schemes, $K = 64$ bins gives each coordinate roughly $\log_2 64 = 6$ bits of resolution per step, sufficient to capture the attractor’s range while keeping the shared vocabulary at a manageable size of 192. K-means uses $K = 512$ clusters in $\mathbb{R}^3$ so that $\log_2 512 = 9$ bits are available per timestep in a single token, matching the total bit budget of the three-token coordinate schemes. Hierarchical quantization uses $K_c = K_r = 32$ so that each two-token pair also provides $2 \log_2 32 = 10$ bits per coordinate while separating coarse and fine structure into distinct token positions. RVQ uses $M = 4$ stages of $K_s = 256$ entries to give $4 \log_2 256 = 32$ bits per timestep, prioritizing reconstruction fidelity at the cost of a longer sequence.

Scheme	Bins / clusters	Tokens / step	Vocab. size [†]
Uniform	64 equal-width bins per coordinate	3	64
Quantile	64 equal-mass bins per coordinate	3	64
K-means	512 clusters in $\mathbb{R}^3$	1	512
Hierarchical	$K_c=32$ coarse, $K_r=32$ fine bins per coordinate	6	192
RVQ	$M=4$ codebook stages, $K_s=256$ entries each	4	256

[†]Vocabulary size is reported per codebook stage for RVQ, and per coordinate for the coordinate-wise schemes (Uniform, Quantile, Hierarchical); the shared token alphabet for those schemes has size $3 \times \text{vocab.}$

Table 1: Summary of the five tokenization schemes used in the ablation study.

Uniform binning. For each coordinate, uniform binning divides the training range into $K = 64$ equal-width intervals. The bin edges are placed at evenly spaced points between the observed minimum and maximum of each coordinate in the training data. A coordinate value v is assigned to the bin whose left edge is the largest edge not exceeding v . The bin index is

$$b_v = \text{clip}\left(K \cdot \frac{v - v_{\min}}{v_{\max} - v_{\min}}, 0, K - 1\right), \quad K = 64.$$

Decoding maps a bin index back to the bin centre, $\hat{v} = \frac{1}{2}(\text{edges}_d[b] + \text{edges}_d[b + 1])$. The three coordinate tokens per timestep are emitted in the order x, y, z , with per-dimension vocabulary offsets $[0, 64, 128]$ so that each coordinate occupies a disjoint range of the shared token alphabet of size $3K = 192$. This scheme is simple and interpretable because the token index has a direct coordinate meaning. Its weakness is that the Lorenz attractor occupies the coordinate bounding box very nonuniformly, so equal-width bins can waste resolution in rarely visited regions.

Quantile binning. Quantile binning replaces equal-width intervals with equal-mass intervals, so that every bin receives approximately the same number of training points. Let

$$q_0 < q_1 < \cdots < q_K$$

be the $K + 1$ empirical quantiles of each coordinate computed from training data, with $K = 64$. A coordinate value is assigned to bin j when $q_j \leq v < q_{j+1}$. If two quantile edges coincide—which

can occur when many training points share the same coordinate value—duplicate quantile edges are adjusted to ensure positive bin widths. Boundary assignment and decoding follow the same procedure as uniform binning. This balances marginal token frequency, but it changes physical scale: dense regions are expanded in token space and sparse regions are compressed. Thus equal token usage does not necessarily mean faithful state-space geometry.

K-means tokenization. K-means tokenization clusters the full standardized state vector $\tilde{\mathbf{x}}_t = (\tilde{x}_t, \tilde{y}_t, \tilde{z}_t)$ in three dimensions. Given $K = 512$ centroids $\{\mu_j\}_{j=1}^K$, the objective is

$$\min_{\{\mu_j\}_{j=1}^K} \sum_i \min_j \|\tilde{\mathbf{x}}_i - \mu_j\|_2^2.$$

Each state is encoded by its nearest centroid,

$$s_t = \arg \min_j \|\tilde{\mathbf{x}}_t - \mu_j\|_2^2,$$

and decoded by returning the centroid vector μ_{s_t} . The tokenizer emits one token per timestep, so the vocabulary size equals the number of clusters.

Unlike coordinate-wise binning, which treats each dimension independently, K-means groups states that are close together in the full three-dimensional space into the same token. This means two states in the same cluster are mapped to identical codes even if they differ slightly, so pointwise reconstruction is coarser. However, because nearby states share the same token, the transition structure the Transformer sees reflects the actual geometry of the attractor rather than an artifact of the coordinate axes.

Hierarchical quantization. A two-level coarse-to-fine scalar quantization scheme represents each coordinate using a coarse code followed by a fine sub-bin index. The first level quantizes the coordinate into $K_c = 32$ equal-width bins, identical in construction to the uniform tokenizer. The second level quantizes the fractional position of the value within its coarse bin into $K_r = 32$ equal-width sub bins, providing sub-bin resolution.

Formally, for coordinate value v in coarse bin b_c with edges $[\ell, h)$, the fractional position is $f = (v - \ell)/(h - \ell) \in [0, 1)$, and the fine bin is $r = \lfloor f \cdot K_r \rfloor$. The pair (b_c, r) together locates v within a sub-interval of width $(h - \ell)/K_r$. Decoding reconstructs

$$\hat{v} = \ell + \frac{r + 0.5}{K_r}(h - \ell),$$

the midpoint of the fine sub-interval. Each coordinate contributes two tokens, giving six tokens per physical timestep in the interleaved order $(c_x, r_x, c_y, r_y, c_z, r_z)$, and a combined vocabulary of size $3K_c + 3K_r = 192$.

Residual vector quantization (RVQ). Residual vector quantization [5] extends the K-means idea by encoding the quantization *error* at each stage rather than discarding it. Given $M = 4$ codebook stages, each with $K_s = 256$ learned centroids, the state $\tilde{\mathbf{x}}_t$ is encoded as follows. The first stage assigns the nearest centroid $\mu_{s_1}^{(1)}$ from codebook $\mathcal{C}^{(1)}$, and the residual $\mathbf{r}^{(1)} = \tilde{\mathbf{x}}_t - \mu_{s_1}^{(1)}$ is passed to the next stage. Each subsequent stage m quantizes the residual from the previous stage:

$$s_m = \arg \min_j \|\mathbf{r}^{(m-1)} - \mu_j^{(m)}\|_2^2, \quad \mathbf{r}^{(m)} = \mathbf{r}^{(m-1)} - \mu_{s_m}^{(m)}.$$

Decoding sums the selected centroids across all stages:

$$\hat{\mathbf{x}}_t = \sum_{m=1}^M \mu_{s_m}^{(m)}.$$

Each timestep is represented by $M = 4$ tokens emitted in stage order (s_1, s_2, s_3, s_4) , with each token drawn from a per-stage vocabulary of size $K_s = 256$. The codebooks are fit jointly on training data by iterating Lloyd’s algorithm [3] on the residuals at each stage. RVQ achieves finer reconstruction than a single-stage quantizer of the same total token budget because successive stages correct the errors of earlier ones, at the cost of producing a longer token sequence whose inter-stage dependencies can be harder for the Transformer to model.

3.3 Autoregressive sequence model

After tokenization, the Transformer is trained as an autoregressive next-token model. Given previous tokens $s_1, \dots, s_t$, it estimates

$$p_{\theta}(s_{t+1} \mid s_1, \dots, s_t).$$

The training loss is cross entropy:

$$\mathcal{J}(\theta) = - \sum_t \log p_{\theta}(s_{t+1} \mid s_1, \dots, s_t).$$

The architecture is shared across tokenizers so that the comparison isolates the representation. The Transformer component should therefore be viewed as a fixed probe of sequence learnability rather than as the main object of optimization. It consists of 4 decoder layers with embedding dimension 128, 8 attention heads, and a feed-forward width of 512, giving roughly 2M parameters. A maximum context length of 512 tokens is used. The model is trained for 25 epochs with the Adam optimizer (learning rate 3×10^{-4} , gradient clipping at 1.0) and a batch size of 64 windows of length 512 sampled with stride 32 within individual trajectories. We note that the learning curves in Figure 5 suggest that K-means and the coordinate-wise schemes reach near-convergence within 25 epochs, while hierarchical quantization and RVQ are still declining at epoch 25. The reported results for those two schemes should therefore be understood as lower bounds on their potential performance; a longer training schedule or larger model capacity might reduce the gap relative to K-means.

4 Evaluation Metrics

The final results use four complementary metrics.

Next-token accuracy. Accuracy is the fraction of predicted next tokens whose top-1 prediction matches the true next token:

$$\text{Acc} = \frac{1}{T} \sum_{t=1}^T \mathbf{1} \left[\arg \max_v p_{\theta}(v \mid s_1, \dots, s_t) = s_{t+1} \right].$$

This is useful as a sanity check, but it is not directly comparable across tokenizers with different numbers of tokens per physical timestep (see $\mathcal{L}$ below).

Reconstruction fidelity. Fidelity is measured by state-space reconstruction RMSE:

$$\mathcal{F} = \sqrt{\frac{1}{N} \sum_{t=1}^N \|\mathbf{x}_t - \hat{\mathbf{x}}_t\|_2^2}, \quad \hat{\mathbf{x}}_t = \text{dec}(\text{enc}(\mathbf{x}_t)).$$

This measures how much continuous state information survives tokenization and decoding.

Normalized learnability. Cross entropy is normalized per physical timestep:

$$\mathcal{L} = \frac{\tau}{\log 2} H(s_{t+1} \mid s_1, \dots, s_t),$$

where τ is the number of tokens emitted per physical timestep and H is held-out cross entropy measured per token in nats. Multiplying per-token nats by τ and dividing by $\log 2$ converts the learning cost to bits per physical timestep, allowing tokenizers with different sequence lengths to be compared on the same scale.

Generative quality. Free-running rollout quality is measured by average one-dimensional Wasserstein distance over coordinate marginals:

$$\mathcal{W} = \frac{1}{3} \sum_{d \in \{x, y, z\}} W_1(\hat{p}_d^{\text{gen}}, p_d^{\text{true}}),$$

where $\hat{p}_d^{\text{gen}}$ is the empirical marginal from a 2,000-step generated rollout and p_d^{true} is the pooled held-out attractor marginal.

5 Results

Figure 2 shows the rank-frequency curves for all five tokenizers. These curves are already informative before examining model performance. Uniform binning has a steep drop-off (log-log slope -1.52) because equal-width coordinate bins interact with the nonuniform density of the Lorenz attractor. Quantile binning is nearly flat by construction (slope -0.00) because each marginal bin receives similar mass. K-means uses a compact set of state-space clusters (slope -0.35), hierarchical tokenization has a moderate slope (-0.60) because its two-stage representation creates a longer stream with repeated coarse and fine codes, and RVQ has a similar moderate slope (-0.66) due to its multi-stage residual structure.

Figure 3 shows per-coordinate reconstruction distributions versus ground truth and vocabulary utilization across schemes. All five tokenizers achieve full vocabulary utilization. K-means achieves the lowest Wasserstein error on all three coordinates by a substantial margin, while hierarchical tokenization performs poorly on the y coordinate in particular. The y coordinate spans a wider and more asymmetric range on the Lorenz attractor than x or z ; because hierarchical quantization applies equal-width bins independently to each coordinate, the coarse level allocates the same number of bins to y as to the other coordinates, leaving its wider tails relatively underresolved. This mismatch between bin resolution and coordinate density is the likely source of the elevated y -coordinate error for the hierarchical scheme.

Figure 4 shows cross-entropy loss and next-token accuracy on both train and test sets, and Figure 5 shows the normalized learnability (bits per physical timestep) over training. K-means converges rapidly to a much lower learnability cost than all other schemes. Hierarchical tokenization begins

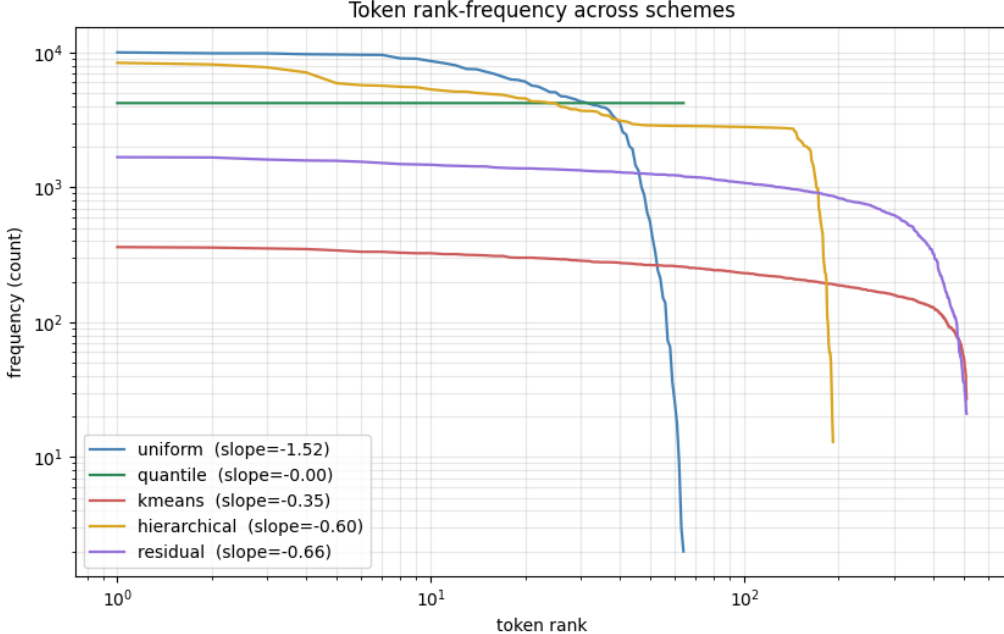


Figure 2: Rank-frequency curves for the five tokenization schemes on a log-log scale.

with a high cost owing to its longer token sequence and converges slowly, remaining well above the coordinate-wise schemes even after 25 epochs. RVQ falls between hierarchical and the coordinate-wise schemes in learnability cost, reflecting the trade-off between its finer reconstruction and its four-token-per-step sequence length.

Table 2 gives the main quantitative comparison across all five schemes. Note that the Acc column is included as a sanity check only; because tokenizers emit different numbers of tokens per physical timestep, raw per-token accuracy is not directly comparable across schemes. All metrics are computed on the held-out test trajectories; because each test trajectory is a single deterministic rollout from a fixed initial condition, variance across rollouts is not reported, but differences of the magnitudes seen in Table 2 are unlikely to be attributable to random variation.

Scheme	$\mathcal{F} \downarrow$ (RMSE)	Acc $\uparrow^*$	$\mathcal{L} \downarrow$ (bits/step)	$\mathcal{W} \downarrow$
Uniform	0.222	0.903	1.4	10.5
Quantile	0.296	0.891	1.6	9.2
K-means	0.468	0.898	0.6	1.0
Hierarchical	0.191	0.859	11.5	11.5
RVQ	0.210	0.871	8.2	7.3

*Acc is not directly comparable across schemes because tokenizers emit different numbers of tokens per physical timestep; $\mathcal{L}$ provides the comparability-corrected measure.

Table 2: Final quantitative results across all five tokenization schemes.

The most direct fidelity result is that hierarchical coding gives the best reconstruction RMSE, $\mathcal{F} = 0.191$. This matches the design of hierarchical quantization: each state receives a coarse approximation followed by a fine sub-bin correction. RVQ achieves the second-lowest RMSE

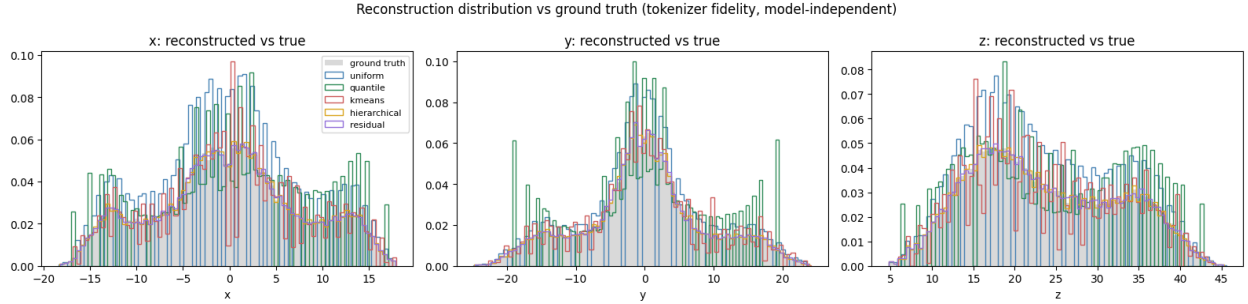


Figure 3: Reconstruction distribution versus ground truth for each tokenization scheme, shown per coordinate.

($\mathcal{F} = 0.210$), since successive codebook stages correct the quantization error of earlier ones. Uniform binning is third in reconstruction, while K-means has the largest RMSE because each full state is decoded as a single cluster centroid rather than a refined estimate. From a pure compression-and-reconstruction perspective, hierarchical coding is therefore the strongest method in the experiment.

The learnability result tells a different story. K-means achieves the lowest normalized cross entropy, well below uniform, quantile, RVQ, or hierarchical tokenization. The key reason is that K-means emits one token per timestep and assigns tokens according to neighborhoods in $\mathbb{R}^3$. The model sees a compact symbolic trajectory through attractor regions rather than three separate coordinate streams or a multi-token correction stream. Hierarchical tokenization pays the steepest learnability cost for its longer sequence: despite encoding more information per timestep, its bits-per-timestep cost is an order of magnitude higher than K-means. RVQ, with four tokens per step and inter-stage dependencies, converges more slowly than the coordinate-wise schemes but does better than hierarchical.

The rollout result supports the same interpretation. K-means obtains the best marginal Wasserstein score by a wide margin, and the qualitative rollout in Figure 6 stays closest to the recognizable two-lobed Lorenz attractor. RVQ performs second on generative quality ($\mathcal{W} = 7.3$), outperforming both coordinate-wise schemes and hierarchical coding. Uniform and quantile rollouts are substantially worse than K-means by $\mathcal{W}$ and visually look less like stable attractor trajectories.

6 Interpretation and Reflections

6.1 Per-Token Accuracy alone is misleading

Uniform binning has the highest per-token accuracy in Table 2 while simultaneously producing the worst free-running rollout quality by a large margin. This discrepancy illustrates why token-level accuracy alone is an unreliable criterion for evaluating tokenizers for dynamical systems. Coordinate-wise uniform bins interact with the local correlation structure of the Lorenz trajectory in a way that makes the next coordinate bin highly predictable over short horizons, supporting strong short-range token accuracy. However, over long rollouts, the generated trajectory is not sufficiently constrained to stay on the attractor, leading the coordinate distribution to drift away from the true measure.

The comparability problem is further compounded by the fact that different tokenizers emit different numbers of tokens per physical timestep. Because uniform and quantile emit three coordinate

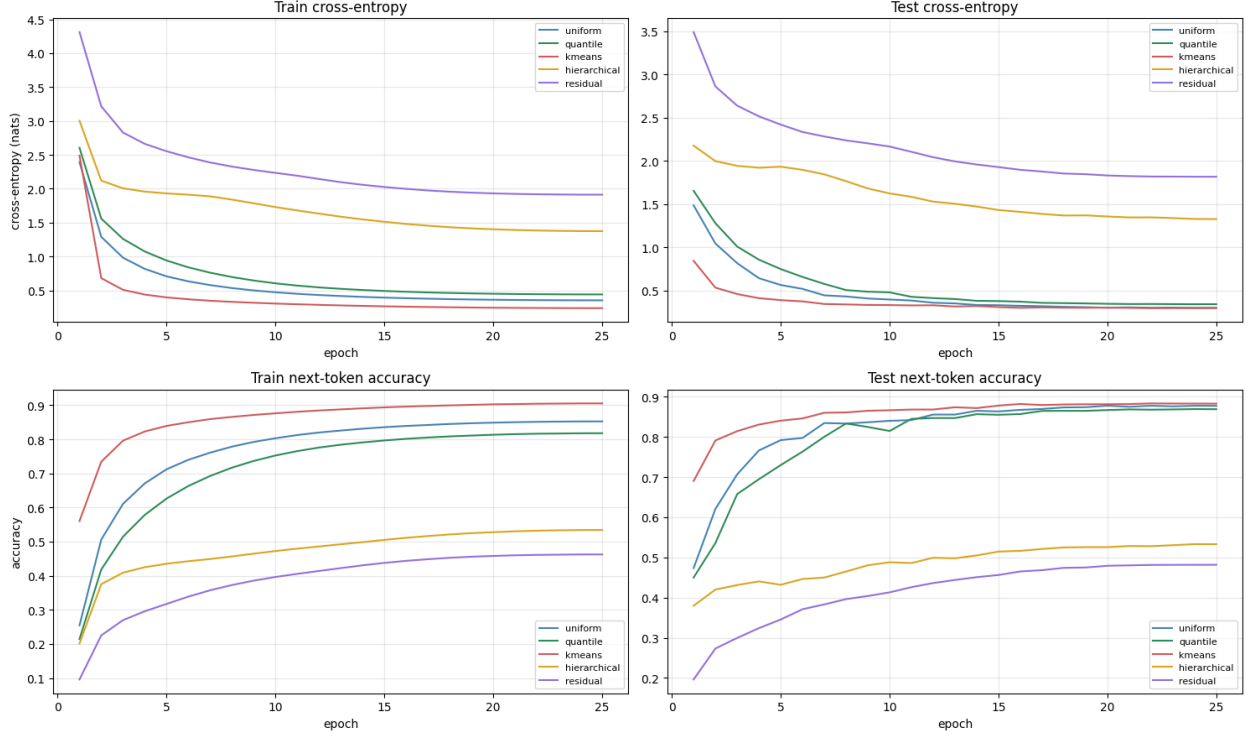


Figure 4: Train and test cross-entropy (top) and next-token accuracy (bottom) across 25 epochs.

tokens per step while K-means emits one, a raw per-token accuracy number does not correspond to the same unit of physical dynamics across conditions. To address this issue, one should consider the normalized learnability metric $\mathcal{L}$, which converts held-out cross-entropy to bits per physical timestep, revealing a gap in learning difficulty hidden by per-token accuracy.

6.2 Reconstruction Fidelity versus Dynamical Learnability

The contrasting profiles of hierarchical quantization, RVQ, and K-means tokenization reveal the central trade-off of the study. Hierarchical quantization and RVQ are both optimized for state reconstruction: hierarchical represents each coordinate at two resolution levels, while RVQ refines the joint-state estimate across four successive codebook stages, together achieving the two lowest decode RMSEs in the experiment. This resolution is precisely what a downstream task requiring accurate state recovery would want.

Conversely, by clustering states according to Euclidean proximity, K-means optimizes for geometric coherence. This leads to tokens being distinguished based on the attractor’s geometry rather than the point distribution. The Transformer then learns to predict transitions between attractor regions rather than transitions between separately coded coordinate values, and this geometric grounding appears to be responsible both for the substantially lower normalized learnability cost and for the stronger generative quality. The lesson is that a representation can be less precise pointwise yet more faithful dynamically, where the attractor geometry provides stronger predictive power above any individual trajectory.

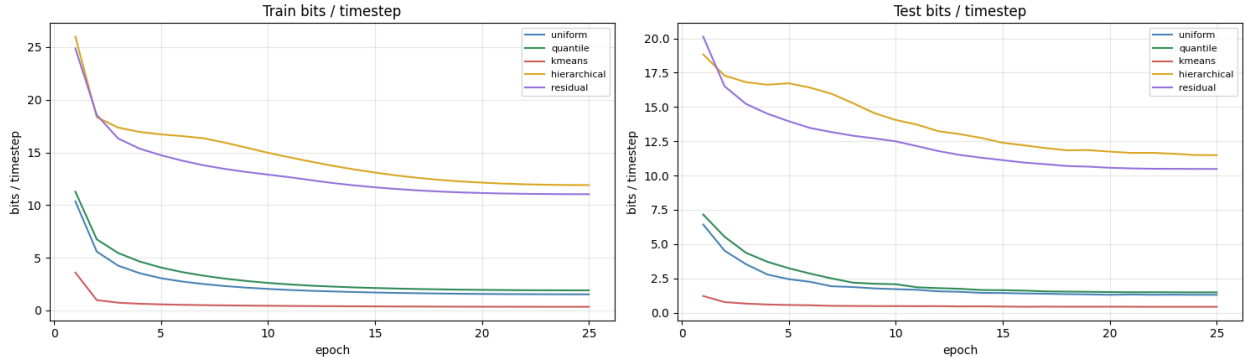


Figure 5: Normalized learnability in bits per physical timestep during training (left) and testing (right) across 25 epochs.

6.3 What quantile binning reveals

Quantile binning offers a useful counter-example to the intuition that uniform token usage is inherently beneficial. By construction, it equalizes the marginal frequency of each bin, which might be expected to aid learning by preventing the model from becoming overconfident on frequently occurring tokens. In practice, however, quantile binning does not rank best on any of the four metrics examined. The reason is that equalizing marginal counts is not the same as preserving state-space geometry: the transformation from physical coordinates to quantile bins stretches dense regions and compresses sparse ones, distorting the metric structure of the attractor in a way that is not corrected simply by increasing vocabulary utilization. This finding reinforces a broader point about tokenizer design: vocabulary utilization is a diagnostic quantity, not an objective. A well-designed tokenizer should allocate resolution according to the scientific structure that matters for downstream use, rather than simply according to the empirical marginal distribution of a single coordinate.

7 Limitations

There are some limitations that should also be pointed out when interpreting these results, the main one being that this project only evaluates one configuration for each tokenizer. Uniform and quantile binning use 64 bins for each coordinate, K-means uses 512 clusters, and RVQ uses $M = 4$ stages with $K_s = 256$ entries each. So these results should not be read as a concluding ranking of tokenizers. They are merely a demonstration of how these current implementations of the tokenizers perform in a single controlled experiment. Changes in vocabulary size could alter the trade-offs between reconstruction, learnability, and rollout quality.

The second limitation is that the Transformer is used as a fixed probe for the token sequences. This is useful for isolating the effect of tokenization, but it also means that a tokenizer might perform worse here because the model capacity, context length, or training schedule may not have been suited for that representation. For example, RVQ maintains more state information, but it also produces a longer, more complex token sequence than K-means. Another model architecture or longer training schedule might affect how hard that sequence is to learn.

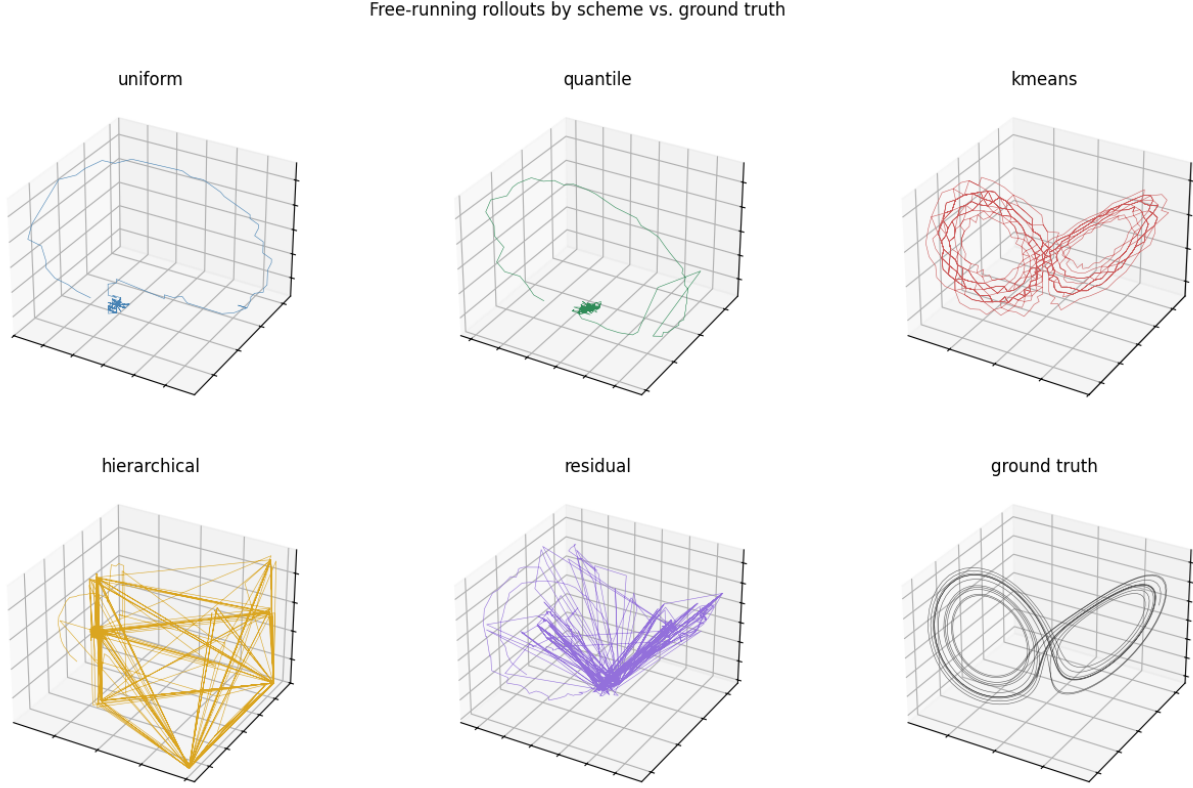


Figure 6: Free-running rollouts by tokenization scheme versus ground truth.

The rollout score is similarly limited. The marginal Wasserstein distance is a good indicator of whether the generated trajectory has approximately correct coordinate marginals, but it is not a thorough test of the Lorenz attractor as a geometric object. A generated rollout could have reasonable x , y , and z marginals and still switch lobes incorrectly, miss the right temporal sequence, or move through state space in an unphysical way. For these reasons, the rollout scores should be viewed as useful but incomplete measures of the generated dynamics.

A few other limitations stem from how the tokenizers are fit. Because all of the bin edges, centroids, and codebooks are learned based on the training trajectories and then applied to the held-out trajectories, test leakage is avoided. However, this means that states outside of the training range or away from well-sampled regions of the attractor must be clipped or assigned to imperfect codes. This is important for coordinate-wise binning, because boundary bins can absorb out-of-range states, and for K-means, because the attractor’s low-density regions may be represented very coarsely.

Finally, this study is limited to the Lorenz system. While Lorenz is a useful test system because it is chaotic, low-dimensional, and visually clear, it is not representative of larger scientific systems. For example, coordinate-wise binning may be infeasible in higher dimensions, K-means clustering may need many more centroids, and RVQ-based compression may need a different codebook architecture. Thus, the findings here should be viewed as results from a controlled Lorenz experiment, not as a general rule for all ODE or PDE systems.

8 Future Work

One thing we never tested in detail is exactly how sensitive our results are to the tokenizer sizes we chose. For this project, each tokenizer was only used once, with 64 bins per coordinate (uniform and quantile binning), 512 clusters (K-means), or a two-level residual codebook (RVQ). This is fine for drawing a clear comparison, but doesn't tell the whole story. An expanded experiment should vary the number of bins, clusters, and residual codes and see how the results vary accordingly. That would tell us if K-means remains the top dynamical representation if we alter the number of clusters, as well as whether RVQ can achieve better rollout performance without making the token sequence too hard for the model.

We also think the rollout diagnostics could be a bit better. The marginal Wasserstein is helpful because it verifies if the generated trajectory is spending approximately the correct proportion of time in the correct coordinate ranges. However, this is only a coarse test. A trajectory can show acceptable x, y, and z marginals and still be traversing the attractor in a physically inconsistent way. As future work, we would compare return maps, lobe switching behavior, autocorrelation, and joint distributions. These would be a more comprehensive verification of whether or not the underlying Lorenz dynamics are well realized.

Trying out a hybrid tokenizer might also be interesting. While the results were quite interesting, they seem to favor K-means and RVQ for different reasons. K-means provides a very compact state space representation and good rollouts, while RVQ preserves much more of the original state information but produces a sequence that is harder to model. A hybrid approach might simply use a state to select a specific K-means region and then store a small residual correction within that region. This would preserve the geometric properties of K-means while recovering some of the information that is eliminated by quantizing a region down to a single centroid.

The final key extension is to go beyond Lorenz. Lorenz is a useful example since the attractor is simple to visualize and is chaotic, but it is still merely three-dimensional. Moving to higher-dimensional systems, such as Kuramoto Sivashinsky dynamics or reduced fluid models, would make tokenization a much more difficult problem. Coordinate binning would likely become impractically inefficient, and K-means and RVQ could also become expensive without more careful codebook design. As the state dimension increases, the codebooks required for K-means and RVQ would need to be more carefully designed.

9 Conclusion

This project demonstrates that tokenization is not merely a neutral step of preprocessing leading up to training a Transformer on a set of ODE trajectories. It transforms the learning problem. We used the same Lorenz data, the same general modeling goal, and the same evaluation settings, but the tokenizer altered the input to the sequence model and what could be learned.

The five tokenizers did not have a single best method. Hierarchical quantization achieved the lowest reconstruction error (best pointwise fidelity through two-level coordinate coding), with RVQ close behind thanks to its multi-stage residual correction. K-means had the worst reconstruction error, which makes sense since all states are decoded as a cluster centroid, but K-means had the best normalized learnability and best rollout score, which was the main result. In a chaotic system, representing the exact state at one time step isn't the same thing as learning a representation that can generate good, long-term dynamics.

Both coordinate-based methods encountered the same problem from a different perspective. Within the token stream, uniform binning gives the most accurate predictions for the next token; in terms of rollout it did not give the same benefit. Quantile binning equalizes the marginal token frequencies, but equal marginal token frequency did not preserve the geometry of the attractor. This makes token accuracy look different entirely, more as a sanity check than an actual performance metric.

For these reasons, measuring how well a tokenizer performs on a dynamical system using a single metric would be misleading. The reconstruction error indicates the loss in information between the state and the symbolic space. The normalized learnability reflects how hard the symbolic sequence is for the Transformer to model per physical timestep. The rollout quality indicates the similarity or divergence in trajectory behavior between the two systems. Missing any one of these three may lead to a false positive or false negative.

From our experiments, we conclude K-means is the most suitable when the objective is to learn a small symbolic representation of Lorenz dynamics and generate plausible rollouts. Though if highly fine-grained reconstruction is required, hierarchical quantization achieves the lowest RMSE, with RVQ offering a favorable intermediate trade-off between reconstruction accuracy and rollout quality. The larger lesson is that the tokenizer should be selected according to the scientific task. For chaotic dynamics, the most suitable representation is not necessarily the one with the best token accuracy or the lowest reconstruction loss, but rather the one whose discrete sequence encodes the geometry and long-run behavior of the system.

References

- [1] E. N. Lorenz. Deterministic nonperiodic flow. *Journal of the Atmospheric Sciences*, 20(2):130–141, 1963. [https://doi.org/10.1175/1520-0469\(1963\)020%3C0130:DNF%3E2.0.CO;2](https://doi.org/10.1175/1520-0469(1963)020%3C0130:DNF%3E2.0.CO;2).
- [2] G. K. Zipf. *Human Behavior and the Principle of Least Effort: An Introduction to Human Ecology*. Addison-Wesley, 1949. Catalog record: <https://www.ncbi.nlm.nih.gov/nlmcatalog/274664>.
- [3] S. P. Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2):129–137, 1982. <https://doi.org/10.1109/TIT.1982.1056489>.
- [4] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017. <https://doi.org/10.48550/arXiv.1706.03762>.
- [5] A. van den Oord, O. Vinyals, and K. Kavukcuoglu. Neural discrete representation learning. *Advances in Neural Information Processing Systems*, 2017. <https://arxiv.org/abs/1711.00937>.